

Khaula Molapo

20240001

Week 2

Artificial Intelligence

1. Scenario Analysis — Intelligent Agent in an IT Application

An intelligent agent is a computer system that can observe its environment, process information, and take actions to achieve a specific goal. A good example is a chatbot used in an IT support system. The chatbot interacts with users through text and receives information about their problems, such as login issues, software errors, or network problems. It analyses the user's request and determines an appropriate response based on its knowledge and programmed rules. For example, if a user reports that they cannot connect to Wi-Fi, the chatbot can provide troubleshooting steps such as checking the network connection, restarting the router, or verifying the password. The chatbot can also ask additional questions when it needs more information. Its environment consists of the users, IT systems, databases, and network services it interacts with. The main goal of the agent is to provide quick and useful technical support while reducing the workload on human IT support staff.

2. Agent Research — Utility-Based Agent

A utility-based agent is an intelligent agent that chooses actions based on how useful or beneficial each possible action is. Instead of simply reacting to a situation, the agent evaluates different choices and selects the one that provides the highest expected utility. Utility represents how desirable an outcome is according to specific goals or preferences. For example, a navigation application can act as a utility-based agent by comparing different routes based on travel time, traffic, distance, and fuel consumption. It can then select the route that provides the best overall result for the user. Utility-based agents are useful when there are several possible actions and no single action is always the best. They allow intelligent systems to make more effective decisions by considering the possible consequences of their actions.

3. Simple Reflex Agent in Python

A simple reflex agent makes decisions using the current situation without considering previous events or future consequences. I created a basic thermostat simulation in Python to demonstrate this concept. The thermostat receives the current room temperature as its input and compares it with a fixed desired temperature. If the room is too cold, the agent turns the heater on. If the room is too hot, the agent turns the heater off. For example, if the desired temperature is 22°C and the current temperature is 18°C, the agent activates the

heater. When the temperature reaches 22°C or higher, it switches the heater off. This activity helped me understand how intelligent agents can sense their environment and respond automatically using predefined rules. However, the agent has limited intelligence because it does not learn from previous situations or consider future conditions.

```
def thermostat_agent(temperature):  
    desired_temperature = 22  
  
    if temperature < desired_temperature:  
        return "Heater ON"  
    else:  
        return "Heater OFF"  
  
temperature = 18  
print(thermostat_agent(temperature))
```

4. Agent Diagram

The agent/environment diagram represents how an intelligent agent interacts with the environment around it. For a chatbot, the chatbot is the agent, while the user, database, network, and other IT systems form part of its environment. The user sends a message to the chatbot, which acts as an input or percept. The chatbot processes the information and determines an appropriate response. It then sends the response back to the user through its interface. The environment can change continuously because users may ask different questions or provide new information. The diagram therefore shows a continuous interaction between the agent and its environment: environment → input/percepts → agent → action/output → environment. This demonstrates that an intelligent agent does not operate completely independently; it observes its environment, makes decisions, and performs actions based on the information it receives.

